

100 Git Troubleshooting Scenarios

Real-World Diagnosis & Fix Guide

DevOps Shack

devopsshack.com | @devopsshack

Quick Scenario Index

#	Title	Category	Pg
1	Undo Last Commit But Keep Changes	Commits & History	—
2	Undo Last Commit and Unstage Changes	Commits & History	—
3	Completely Discard Last Commit and Changes	Commits & History	—
4	Wrong Commit Message — Amend Before Push	Commits & History	—
5	Committed to Wrong Branch	Commits & History	—
6	Accidentally Committed Sensitive Data (Password/Key)	Commits & History	—
7	Detached HEAD State	Commits & History	—
8	Lost Commits After git reset --hard	Commits & History	—
9	Revert a Pushed Commit Safely	Commits & History	—
10	Edit a Commit Deep in History	Commits & History	—
11	Squash Multiple Commits Into One	Commits & History	—
12	Cherry-Pick Commit to Another Branch	Commits & History	—
13	Commit Shows Wrong Author Name/Email	Commits & History	—
14	Empty Commit Created Accidentally	Commits & History	—
15	Merge Conflict — Cannot Auto-Merge	Branching & Merging	—
16	Accidentally Deleted Local Branch	Branching & Merging	—
17	Branch Diverged From Remote — Cannot Fast-Forward	Branching & Merging	—
18	Rebase Conflict During git pull --rebase	Branching & Merging	—
19	Merge Created Unnecessary Merge Commit	Branching & Merging	—
20	Feature Branch Out of Date With Main	Branching & Merging	—
21	Cannot Delete Remote Branch	Branching & Merging	—
22	Stale Remote-Tracking Branches Cluttering List	Branching & Merging	—
23	Wrong Base Branch for Rebase	Branching & Merging	—
24	Merge Commit Reverting Wanted — Then Re-applying Feature	Branching & Merging	—
25	Branch Protection Preventing Direct Push to Main	Branching & Merging	—
26	Fast-Forward Only Merge Failing	Branching & Merging	—
27	Octopus Merge Failing Due to Conflicts	Branching & Merging	—
28	Branch Name With Special Characters Causing Issues	Branching & Merging	—
29	Push Rejected — Remote Has Changes Not in Local	Remote & Push/Pull	—

#	Title	Category	Pg
30	Force Push Accidentally Overwrote Team's Commits	Remote & Push/Pull	—
31	Cannot Push — Authentication Failure		—
32	Wrong Remote URL — Pushing to Wrong Repository		—
33	Shallow Clone Missing History		—
34	Large File Rejected by Remote		—
35	Remote Origin Not Set		—
36	Upstream Not Set — No Tracking Branch		—
37	git clone Fails — Repository Not Found	Remote & Push/Pull	—
38	Fetch Not Updating Remote-Tracking Branches	Remote & Push/Pull	—
39	Push Times Out on Large Repository	Remote & Push/Pull	—
40	Submodule Not Cloned After git clone	Remote & Push/Pull	—
41	Submodule Pointing to Wrong Commit		—
42	Multiple Remotes — Pushing to Wrong One		—
43	Accidentally Staged Wrong Files	Staging & Working Directory	—
44	Discard All Uncommitted Changes		—
45	Stash Not Applying Correctly		—
46	Stash Lost After git stash drop		—
47	Untracked Files Blocking Operations		—
48	Partial File Staging — git add -p		—
49	Line Ending Conflicts (CRLF vs LF)		—
50	File Mode Changes Showing as Modified		—
51	git clean Removed Untracked Files Accidentally		—
52	.gitignore Not Ignoring Already-Tracked File		—
53	git diff Shows Nothing Despite Modified Files		—
54	Wrong File Deleted — Restore From Last Commit		—
55	Tag Not Pushed to Remote	Tags & Releases	—
56	Wrong Tag on Wrong Commit		—
57	Annotated vs Lightweight Tag Confusion		—
58	Cannot Delete Remote Tag		—
59	Tag Not Triggering CI Pipeline		—
60	Listing Tags in Version Order	Tags & Releases	—
61	Global git Config Not Applied to Repo	Configuration & Setup	—
62	Wrong Default Editor Opening for git commit		—
63	GPG Signing Commits Failing		—

#	Title	Category	Pg
64	SSH Config Not Using Correct Key for Host	Configuration & Setup	—
65	git aliases Not Working	Configuration & Setup	—
66	git log Showing Garbled Unicode Characters	Configuration & Setup	—
67	Credential Helper Not Caching Passwords	Configuration & Setup	—
68	core.hooksPath Not Loading Custom Hooks	Configuration & Setup	—
69	diff.tool Not Opening Configured Diff Tool	Configuration & Setup	—
70	init.defaultBranch Still Creating 'master'	Configuration & Setup	—
71	CI Failing on Linting Due to Line Endings	CI/CD & Workflows	—
72	Merge Queue / Protected Branch Blocking PR	CI/CD & Workflows	—
73	git bisect Finding Regression	CI/CD & Workflows	—
74	Pre-commit Hook Blocking All Commits	CI/CD & Workflows	—
75	Repository Too Large — Slow Clone in CI	CI/CD & Workflows	—
76	Commit Message Convention Failing CI Check	CI/CD & Workflows	—
77	GitHub Actions Token Lacks Push Permission	CI/CD & Workflows	—
78	Monorepo — CI Triggering on Unrelated Changes	CI/CD & Workflows	—
79	git worktree Causing Unexpected Conflicts	CI/CD & Workflows	—
80	Changelog Not Generated Correctly From Commits	CI/CD & Workflows	—
81	git filter-repo Removing File From Entire History	Advanced Git	—
82	Repository Corruption — Object Database Error	Advanced Git	—
83	git gc Making Repository Slower Temporarily	Advanced Git	—
84	Partial Clone Not Fetching Needed Objects	Advanced Git	—
85	git rebase -i Accidentally Dropping Commits	Advanced Git	—
86	Merge Strategy — Ours vs Theirs Confusion	Advanced Git	—
87	git notes Not Preserved After Rebase	Advanced Git	—
88	Alternates Object Store Broken	Advanced Git	—
89	git archive Missing Files	Advanced Git	—
90	Ref Namespace Collision	Advanced Git	—
91	Broken git Hooks After Repository Clone	Advanced Git	—
92	Repository Migrated — Refs Not Updated	Advanced Git	—
93	git log --all Not Showing Expected Branch	Advanced Git	—
94	Pack File Corruption After Interrupted Push	Advanced Git	—
95	git sparse-checkout Not Excluding Directories	Advanced Git	—
96	Bundle File Cannot Be Imported	Advanced Git	—
97	git replace Causing Confusing History	Advanced Git	—

#	Title	Category	Pg
98	Shallow Clone Causing rebase/bisect Failures	Advanced Git	—
99	Repository Growing Too Large — Identifying Bloat	Advanced Git	—
100	git maintenance Not Running Automatically	Advanced Git	—

COMMITTS & HISTORY — Real-World Troubleshooting

#1 Undo Last Commit But Keep Changes

SYMPTOM	Committed too early or with wrong message; need to undo commit but keep file changes staged.
ROOT CAUSE	Premature commit before changes were complete or properly reviewed.
FIX	Use git reset --soft HEAD~1 to undo commit but keep changes staged. Edit and recommit.
<pre>git reset --soft HEAD~1</pre>	

#2 Undo Last Commit and Unstage Changes

SYMPTOM	Need to undo last commit and unstage all changes (back to working directory).
ROOT CAUSE	Committed wrong files or wrong state; need full redo.
FIX	git reset --mixed HEAD~1 moves changes back to working directory, unstaged.
<pre>git reset --mixed HEAD~1</pre>	

#3 Completely Discard Last Commit and Changes

SYMPTOM	Last commit is completely wrong and all changes should be discarded permanently.
ROOT CAUSE	Wrong branch committed; accidental commit that needs full removal.
FIX	git reset --hard HEAD~1 destroys the commit and all changes. Cannot be undone easily.
<pre>git reset --hard HEAD~1</pre>	

#4 Wrong Commit Message — Amend Before Push

SYMPTOM	Typo or unclear message in last commit; not yet pushed to remote.
ROOT CAUSE	Commit message written hastily or with error.
FIX	git commit --amend -m 'New correct message' rewrites the last commit message.
<pre>git commit --amend -m "fix: correct commit message"</pre>	

#5 Committed to Wrong Branch

SYMPTOM	Commit was made on main instead of feature branch or vice versa.
ROOT CAUSE	Forgot to create/switch branch before committing.

FIX	Create correct branch at current position, then reset main back. Use cherry-pick if further back.
	<pre>git branch feature/correct-branch git reset --hard HEAD~1 git checkout feature/correct-branch</pre>

#6 Accidentally Committed Sensitive Data (Password/Key)

SYMPTOM	API key, password, or secret accidentally committed to repository history.
ROOT CAUSE	Hardcoded secret in file that was committed; .gitignore not excluding secrets file.
FIX	Immediately rotate the secret. Use git filter-repo to remove from history. Force push.
	<pre>git filter-repo --path secrets.txt --invert-paths git push --force-with-lease</pre>

#7 Detached HEAD State

SYMPTOM	git status shows HEAD detached at <commit>; commits not on any branch.
ROOT CAUSE	Checked out a specific commit hash, tag, or remote branch directly instead of a local branch.
FIX	Create a branch to save work: git checkout -b new-branch. Or return: git checkout main.
	<pre>git checkout -b my-recovery-branch # OR discard and return: git checkout main</pre>

#8 Lost Commits After git reset --hard

SYMPTOM	Accidentally ran git reset --hard; commits appear lost.
ROOT CAUSE	Hard reset discards commits from current branch tip but they remain in reflog.
FIX	Use git reflog to find lost commit SHA, then git checkout or cherry-pick it.
	<pre>git reflog git checkout -b recovery-branch <lost-commit-sha></pre>

#9 Revert a Pushed Commit Safely

SYMPTOM	Bad commit pushed to shared branch; need to undo without rewriting history.
ROOT CAUSE	Cannot force-push shared branch; need a safe undo that preserves history.
FIX	git revert creates a new commit that undoes the target commit. Safe for shared branches.
	<pre>git revert <commit-sha> git push origin main</pre>

#10 Edit a Commit Deep in History

SYMPTOM	Need to fix a commit message or content several commits back, not just the last one.
ROOT CAUSE	Bug or typo discovered in older commit; needs cleanup before PR merge.
FIX	git rebase -i HEAD~N to interactive rebase. Mark target commit as 'reword' or 'edit'.
<pre>git rebase -i HEAD~5 # Mark commit as 'reword' or 'edit' in editor</pre>	

#11 Squash Multiple Commits Into One

SYMPTOM	Feature branch has 15 small messy commits; want to squash into 1 clean commit for PR.
ROOT CAUSE	Incremental development creates noisy history that should be cleaned before merge.
FIX	Interactive rebase: git rebase -i HEAD~N. Mark all but first as 'squash' or 'fixup'.
<pre>git rebase -i HEAD~15 # Mark commits 2-15 as 'squash' or 'fixup'</pre>	

#12 Cherry-Pick Commit to Another Branch

SYMPTOM	Need to apply a specific bug-fix commit from one branch to another without merging entire branch.
ROOT CAUSE	Hotfix needed on main but was made on dev branch; or selective backport required.
FIX	git cherry-pick <sha> applies the diff of that specific commit to current branch.
<pre>git checkout main git cherry-pick <commit-sha></pre>	

#13 Commit Shows Wrong Author Name/Email

SYMPTOM	git log shows commits with wrong name or email; contributor credits incorrect.
ROOT CAUSE	git config user.name/user.email not set globally or per-repo before committing.
FIX	Set correct config. Use git commit --amend for last commit. Use filter-repo for bulk fix.
<pre>git config --global user.name "Your Name" git config --global user.email "you@example.com" git commit --amend --reset-author</pre>	

#14 Empty Commit Created Accidentally

SYMPTOM	git log shows commit with no file changes; useless empty entry in history.
ROOT CAUSE	git commit run without staged changes and --allow-empty used by mistake.
FIX	Remove with interactive rebase and 'drop', or amend if it's the last commit.
<pre>git rebase -i HEAD~2 # Mark empty commit as 'drop'</pre>	

BRANCHING & MERGING — Real-World Troubleshooting

#15 Merge Conflict — Cannot Auto-Merge

SYMPTOM	git merge or git pull fails with CONFLICT message; automatic merge failed.
ROOT CAUSE	Same lines edited in both branches; Git cannot determine which version to keep.
FIX	Open conflicted files, resolve <<<< markers manually. Then git add and git commit.
	<pre>git status # see conflicted files git mergetool git add <resolved-files> git commit</pre>

#16 Accidentally Deleted Local Branch

SYMPTOM	git branch -d run on wrong branch; local branch with unmerged work deleted.
ROOT CAUSE	Typo in branch name or premature deletion before merge.
FIX	git reflog to find last commit of deleted branch. Recreate with checkout -b.
	<pre>git reflog --all grep <branch-name> git checkout -b <branch-name> <sha></pre>

#17 Branch Diverged From Remote — Cannot Fast-Forward

SYMPTOM	git pull fails: Your branch and origin/main have diverged.
ROOT CAUSE	Local and remote both have commits the other doesn't; simple pull cannot fast-forward.
FIX	git pull --rebase to replay local commits on top of remote, or git merge to create merge commit.
	<pre>git pull --rebase origin main # OR git fetch origin && git merge origin/main</pre>

#18 Rebase Conflict During git pull --rebase

SYMPTOM	git pull --rebase stops mid-way with conflict; unclear how to continue.
ROOT CAUSE	Local commits conflict with incoming remote commits during rebase replay.
FIX	Resolve conflicts in each file. git add resolved files. git rebase --continue.
	<pre># resolve conflicts git add <resolved-file> git rebase --continue # OR abort: git rebase --abort</pre>

#19 Merge Created Unnecessary Merge Commit

SYMPTOM	git log --graph shows messy merge commits everywhere; history hard to read.
---------	---

ROOT CAUSE	git pull defaults to merge strategy creating merge commits on every pull.
FIX	Set pull.rebase=true globally to default to rebase on pull for cleaner history.
	<pre>git config --global pull.rebase true # Undo last merge commit: git reset --hard HEAD~1</pre>

#20 Feature Branch Out of Date With Main

SYMPTOM	Feature branch is many commits behind main; need latest changes without merge commit.
ROOT CAUSE	Long-running feature branch not kept up to date with main development.
FIX	git rebase main while on feature branch to replay feature commits on top of latest main.
	<pre>git checkout feature/my-feature git rebase main</pre>

#21 Cannot Delete Remote Branch

SYMPTOM	git push origin --delete <branch> returns error: remote ref does not exist.
ROOT CAUSE	Branch already deleted remotely, or name typo, or insufficient permissions.
FIX	git fetch --prune to sync remote refs. Check branch exists: git branch -r.
	<pre>git fetch --prune git branch -r grep <branch> git push origin --delete <branch-name></pre>

#22 Stale Remote-Tracking Branches Cluttering List

SYMPTOM	git branch -r shows many branches that no longer exist on remote.
ROOT CAUSE	Remote branches deleted but local tracking refs not pruned.
FIX	git fetch --prune removes stale remote-tracking refs. Set fetch.prune=true to auto-prune.
	<pre>git fetch --prune git config --global fetch.prune true git remote prune origin</pre>

#23 Wrong Base Branch for Rebase

SYMPTOM	git rebase target-branch rebased onto wrong branch; commit history is messy.
ROOT CAUSE	Wrong branch checked out or wrong target specified for rebase.
FIX	git rebase --abort to stop. Use git rebase --onto for precise rebase control.
	<pre>git rebase --abort # Rebase correctly: git rebase --onto <new-base> <old-base> <branch></pre>

#24 Merge Commit Reverting Wanted — Then Re-applying Feature

SYMPTOM	Reverted a merge commit but now need to re-apply the feature.
ROOT CAUSE	Reverting a merge removes changes but the re-merge won't reintroduce them since Git sees them as already applied.
FIX	Revert the revert commit (double revert) before re-merging the feature branch.
<pre>git revert <revert-commit-sha> git merge feature/branch</pre>	

#25 Branch Protection Preventing Direct Push to Main

SYMPTOM	git push to main fails: protected branch hook declined.
ROOT CAUSE	Branch protection rules require PR/MR; direct pushes blocked by server.
FIX	Create feature branch, push there, open PR. Or admin bypasses protection temporarily.
<pre>git checkout -b fix/my-change git push origin fix/my-change # Then open Pull Request</pre>	

#26 Fast-Forward Only Merge Failing

SYMPTOM	git merge --ff-only fails: Not possible to fast-forward, aborting.
ROOT CAUSE	Target branch has diverged from source; fast-forward impossible without rebase.
FIX	Rebase feature branch onto target first, then fast-forward merge.
<pre>git rebase main git checkout main git merge --ff-only feature/branch</pre>	

#27 Octopus Merge Failing Due to Conflicts

SYMPTOM	Merging 3+ branches simultaneously fails with conflict during octopus merge.
ROOT CAUSE	Octopus merge (multiple parents) cannot handle conflicting changes across branches.
FIX	Merge branches sequentially one at a time, resolving conflicts at each step.
<pre>git merge branch1 git merge branch2 git merge branch3</pre>	

#28 Branch Name With Special Characters Causing Issues

SYMPTOM	Branch names with slashes or special chars cause issues in scripts or CI.
ROOT CAUSE	Branch names like feature/my-feat or fix/issue#123 cause parsing problems in some tools.
FIX	Rename branch: git branch -m old-name new-name. Avoid #, spaces, backslashes in names.
<pre>git branch -m 'fix/issue#123' 'fix/issue-123' git push origin :fix/issue#123 fix/issue-123</pre>	

REMOTE & PUSH/PULL — Real-World Troubleshooting

#29 Push Rejected — Remote Has Changes Not in Local

SYMPTOM	git push fails: Updates were rejected because the remote contains work you do not have.
ROOT CAUSE	Someone else pushed to remote after your last pull; histories diverged.
FIX	git pull --rebase to get remote changes, then push. Never force push shared branches.
	<pre>git pull --rebase origin main git push origin main</pre>

#30 Force Push Accidentally Overwrote Team's Commits

SYMPTOM	After git push --force, team members' commits disappeared from remote.
ROOT CAUSE	git push --force replaces remote history with local history, discarding remote-only commits.
FIX	Use --force-with-lease instead of --force. Restore from team member's local clone.
	<pre># Safer force push: git push --force-with-lease origin main # Restore overwritten: git push origin <teammate-sha>:main</pre>

#31 Cannot Push — Authentication Failure

SYMPTOM	git push returns: Authentication failed or Permission denied (publickey).
ROOT CAUSE	SSH key not added to agent or GitHub/GitLab, expired token, or wrong remote URL protocol.
FIX	Check SSH: ssh -T git@github.com. For HTTPS, update credentials or use PAT.
	<pre>ssh -T git@github.com git remote set-url origin git@github.com:user/repo.git ssh-add ~/.ssh/id_ed25519</pre>

#32 Wrong Remote URL — Pushing to Wrong Repository

SYMPTOM	git push succeeds but changes not in expected repository; pushed to fork instead of upstream.
ROOT CAUSE	Remote origin pointing to fork; upstream remote not configured correctly.
FIX	Check remotes: git remote -v. Update URL: git remote set-url.
	<pre>git remote -v git remote set-url origin git@github.com:org/correct-repo.git</pre>

#33 Shallow Clone Missing History

SYMPTOM	git log shows only limited commits; git blame or bisect fails with missing history.
ROOT CAUSE	Repository cloned with --depth flag (shallow clone); full history not downloaded.
FIX	Fetch full history: git fetch --unshallow.
	<pre>git fetch --unshallow git log --oneline wc -l</pre>

#34 Large File Rejected by Remote

SYMPTOM	git push fails: File <name> is 150.00 MB; exceeds GitHub's 100MB file size limit.
ROOT CAUSE	Large binary, dataset, or compiled artifact accidentally committed to repository.
FIX	Remove with git filter-repo. Use Git LFS for large files going forward.
	<pre>git filter-repo --path large-file.bin --invert-paths git lfs track '*.bin' git push --force-with-lease</pre>

#35 Remote Origin Not Set

SYMPTOM	git push fails: No configured push destination. fatal: 'origin' does not appear to be a git repository.
ROOT CAUSE	Repository initialized locally but no remote added yet.
FIX	Add remote origin: git remote add origin <url>. Then set upstream.
	<pre>git remote add origin git@github.com:user/repo.git git push -u origin main</pre>

#36 Upstream Not Set — No Tracking Branch

SYMPTOM	git pull fails: There is no tracking information for the current branch.
ROOT CAUSE	Local branch has no upstream remote branch configured.
FIX	Set upstream: git push -u origin <branch> or git branch --set-upstream-to.
	<pre>git push -u origin feature/my-branch # OR git branch --set-upstream-to=origin/main main</pre>

#37 git clone Fails — Repository Not Found

SYMPTOM	git clone returns: Repository not found or Could not read from remote repository.
ROOT CAUSE	Wrong URL, private repo without access, or SSH key not authorised for that repo.
FIX	Verify URL in GitHub/GitLab UI. Check SSH access. Use HTTPS with PAT if SSH fails.
	<pre>git clone https://github.com/user/repo.git # Verify SSH: ssh -T git@github.com</pre>

#38 Fetch Not Updating Remote-Tracking Branches

SYMPTOM	git fetch runs without error but git branch -r shows outdated branches.
ROOT CAUSE	Fetch refspec not configured to update all remote branches, or wrong remote specified.
FIX	Ensure refspec covers all branches: git fetch --all. Check remote config.
	<pre>git fetch --all --prune git remote show origin</pre>

#39 Push Times Out on Large Repository

SYMPTOM	git push hangs or times out when pushing large number of objects.
ROOT CAUSE	Network timeout, server-side post-receive hook slow, or pack size too large.
FIX	Increase HTTP buffer: git config http.postBuffer 524288000. Push in smaller batches.
	<pre>git config --global http.postBuffer 524288000 git config --global http.lowSpeedLimit 0 git config --global http.lowSpeedTime 999999</pre>

#40 Submodule Not Cloned After git clone

SYMPTOM	Directory for submodule exists but is empty after cloning parent repo.
ROOT CAUSE	Submodules not initialized/updated; git clone doesn't fetch submodules by default.
FIX	Run git submodule update --init --recursive. Or clone with --recurse-submodules.
	<pre>git submodule update --init --recursive # OR clone with: git clone --recurse-submodules <url></pre>

#41 Submodule Pointing to Wrong Commit

SYMPTOM	Submodule shows modified state; CI fails because submodule is on unexpected commit.
ROOT CAUSE	Submodule commit pointer in parent repo not updated after submodule changed.
FIX	Update submodule to latest: git submodule update --remote. Commit the pointer update.
	<pre>git submodule update --remote --merge git add <submodule-path> git commit -m "chore: update submodule to latest"</pre>

#42 Multiple Remotes — Pushing to Wrong One

SYMPTOM	git push sends to origin (fork) instead of upstream, or vice versa.
ROOT CAUSE	Multiple remotes configured; default push target is unclear.
FIX	Specify remote explicitly: git push upstream main. Check: git remote -v.
	<pre>git remote -v git push upstream main git push origin feature/branch</pre>

STAGING & WORKING DIRECTORY — Real-World Troubleshooting

#43 Accidentally Staged Wrong Files

SYMPTOM	git add . staged files that shouldn't be committed (config, secrets, build artifacts).
ROOT CAUSE	Glob pattern too broad; .gitignore not configured properly.
FIX	git reset HEAD <file> or git restore --staged <file> to unstage specific files.
	<pre>git restore --staged config/local.yml git restore --staged .</pre>

#44 Discard All Uncommitted Changes

SYMPTOM	Working directory has unwanted changes; need to revert all files to last commit state.
ROOT CAUSE	Experimental changes, wrong edits, or corruption that should be thrown away.
FIX	git restore . or git checkout -- . to discard all working directory changes.
	<pre>git restore . # OR older syntax: git checkout -- .</pre>

#45 Stash Not Applying Correctly

SYMPTOM	git stash pop fails with conflict or stash doesn't restore expected files.
ROOT CAUSE	Stashed changes conflict with current branch state, or stash was created on different branch.
FIX	git stash list to view stashes. git stash show -p stash@{0} to inspect. Resolve conflicts.
	<pre>git stash list git stash show -p stash@{0} git stash apply stash@{0}</pre>

#46 Stash Lost After git stash drop

SYMPTOM	Accidentally dropped a stash with important changes.
ROOT CAUSE	git stash drop or git stash pop removes the stash entry.
FIX	Stash commits remain in reflog briefly. git fsck --unreachable to find dangling commits.
	<pre>git fsck --unreachable grep commit awk '{print \$3}' xargs git log --oneline --no-walk</pre>

#47 Untracked Files Blocking Operations

SYMPTOM	git checkout or git merge fails: Your local changes would be overwritten.
ROOT CAUSE	Untracked or modified files in working directory conflict with incoming changes.

FIX	Stash changes: git stash. Or discard: git clean -fd for untracked, git restore . for modified.
	<pre>git stash push -u # stash including untracked # OR discard: git clean -fd && git restore .</pre>

#48 Partial File Staging — git add -p

SYMPTOM	File has both good and bad changes; want to stage only specific lines/hunks.
ROOT CAUSE	Mixed changes in one file; committing entire file would include unwanted changes.
FIX	git add -p for interactive hunk-by-hunk staging. Press y/n/s to split hunks.
	<pre>git add -p <filename> # y = stage hunk, n = skip, s = split hunk, e = edit manually</pre>

#49 Line Ending Conflicts (CRLF vs LF)

SYMPTOM	Every file shows as modified; git diff shows ^M characters; Windows/Linux mismatch.
ROOT CAUSE	core.autocrlf setting differs between team members; Windows adds CR to line endings.
FIX	Set core.autocrlf=input on Mac/Linux; true on Windows. Add .gitattributes for consistency.
	<pre>git config --global core.autocrlf input echo '* text=auto' > .gitattributes git add --renormalize .</pre>

#50 File Mode Changes Showing as Modified

SYMPTOM	git status shows files as modified but content is identical; diff shows mode change.
ROOT CAUSE	File execute permissions changed (chmod); core.fileMode=true tracking the change.
FIX	If mode changes are unintentional: git config core.fileMode false. Or restore correct modes.
	<pre>git config core.fileMode false # OR fix mode: git diff --stat git restore <file></pre>

#51 git clean Removed Untracked Files Accidentally

SYMPTOM	git clean -fd deleted untracked files that were needed.
ROOT CAUSE	git clean permanently removes untracked files; -f flag bypasses confirmation.
FIX	Always use -n (dry-run) first: git clean -nfd. If files deleted, may recover from editor/IDE history.
	<pre>git clean -nfd # dry run first git clean -fd # actual clean</pre>

#52 .gitignore Not Ignoring Already-Tracked File

SYMPTOM	File added to .gitignore still appears in git status as modified.
ROOT CAUSE	File was previously committed; .gitignore only affects untracked files.
FIX	Remove from tracking: <code>git rm --cached <file></code> . Then commit the .gitignore change.
<pre>git rm --cached config/local.yml git add .gitignore git commit -m "chore: untrack local config"</pre>	

#53 git diff Shows Nothing Despite Modified Files

SYMPTOM	Files visually changed but git diff shows no output.
ROOT CAUSE	Changes already staged (use <code>git diff --staged</code>) or file permissions only changed.
FIX	<code>git diff --staged</code> to see staged changes. <code>git diff HEAD</code> to see all changes vs last commit.
<pre>git diff --staged git diff HEAD git status -s</pre>	

#54 Wrong File Deleted — Restore From Last Commit

SYMPTOM	<code>git rm</code> or manual deletion removed a tracked file that is still needed.
ROOT CAUSE	Accidental file removal committed or staged.
FIX	<code>git restore <file></code> to recover from last commit. If committed, use <code>git show HEAD:<file></code> .
<pre>git restore <deleted-file> # If already committed: git checkout HEAD~1 - <deleted-file></pre>	

TAGS & RELEASES — Real-World Troubleshooting

#55 Tag Not Pushed to Remote

SYMPTOM	Local tag created but not visible in GitHub/GitLab releases; CI tag trigger not firing.
ROOT CAUSE	git push doesn't push tags by default; tags must be pushed explicitly.
FIX	git push origin <tagname> or git push --tags to push all tags.
	<pre>git push origin v1.2.3 # OR push all tags: git push --tags</pre>

#56 Wrong Tag on Wrong Commit

SYMPTOM	Release tag v2.0.0 applied to wrong commit; release points to old or incorrect code.
ROOT CAUSE	Tag created while on wrong branch or wrong commit checked out.
FIX	Delete wrong tag locally and remotely. Recreate on correct commit.
	<pre>git tag -d v2.0.0 git push origin :refs/tags/v2.0.0 git tag v2.0.0 <correct-commit-sha> git push origin v2.0.0</pre>

#57 Annotated vs Lightweight Tag Confusion

SYMPTOM	Tag created but missing tagger info, date, or message in GitHub releases.
ROOT CAUSE	Lightweight tag used instead of annotated tag; lightweight tags have no metadata.
FIX	Use git tag -a for annotated tags with message. Delete lightweight and recreate annotated.
	<pre>git tag -a v1.0.0 -m "Release version 1.0.0" git push origin v1.0.0</pre>

#58 Cannot Delete Remote Tag

SYMPTOM	git push origin --delete <tag> fails or tag reappears on remote.
ROOT CAUSE	Tag protected, permission issue, or wrong refspec syntax.
FIX	Use full refspec: git push origin :refs/tags/<tagname>.
	<pre>git push origin :refs/tags/v1.0.0-bad git tag -d v1.0.0-bad</pre>

#59 Tag Not Triggering CI Pipeline

SYMPTOM	New tag pushed but CI/CD pipeline for release not triggered.
ROOT CAUSE	CI config not matching tag pattern, or pipeline only triggers on branch pushes.

FIX	Check CI config (on: push: tags: pattern). Ensure tag follows expected pattern (v*).
	<pre># GitHub Actions example: # on: # push: # tags: # - 'v*' git tag v1.0.1 && git push origin v1.0.1</pre>

#60 Listing Tags in Version Order

SYMPTOM	git tag output shows tags in alphabetical order; v1.10.0 listed before v1.9.0.
ROOT CAUSE	Default tag listing is lexicographic; semantic versioning doesn't sort alphabetically.
FIX	Use git tag --sort=version:refname for semantic version sorting.
	<pre>git tag --sort=version:refname git tag --sort=-version:refname # descending</pre>

CONFIGURATION & SETUP — Real-World Troubleshooting

#61 Global git Config Not Applied to Repo

SYMPTOM	Global config settings (user.name, core.editor) not taking effect in specific repo.
ROOT CAUSE	Local repo config overrides global; a local git config --local entry exists.
FIX	Check local config: git config --local --list. Remove overrides or set correct values.
<pre>git config --local --list git config --local --unset user.email git config --list --show-origin</pre>	

#62 Wrong Default Editor Opening for git commit

SYMPTOM	git commit opens nano/vim/notepad instead of preferred editor.
ROOT CAUSE	core.editor not set in git config; inherits EDITOR environment variable or OS default.
FIX	Set preferred editor: git config --global core.editor 'code --wait'.
<pre>git config --global core.editor 'code --wait' # For vim: git config -- global core.editor vim</pre>	

#63 GPG Signing Commits Failing

SYMPTOM	git commit -S fails: error: gpg failed to sign the data.
ROOT CAUSE	GPG key not configured, expired, or GPG agent not running.
FIX	Set GPG key: git config --global user.signingkey <KEYID>. Start gpg-agent.
<pre>gpg --list-secret-keys --keyid-format LONG git config --global user.signingkey <KEY_ID> git config --global commit.gpgsign true</pre>	

#64 SSH Config Not Using Correct Key for Host

SYMPTOM	git push uses wrong SSH key; multiple GitHub accounts or different keys per host.
ROOT CAUSE	Default SSH key used instead of per-host configured key in ~/.ssh/config.
FIX	Configure ~/.ssh/config with Host blocks specifying IdentityFile per hostname.
<pre># ~/.ssh/config: # Host github-work # HostName github.com # User git # IdentityFile ~/.ssh/id_ed25519_work git remote set-url origin git@github-work:org/repo.git</pre>	

#65 git aliases Not Working

SYMPTOM	Custom git alias (e.g., git st) returns: git: 'st' is not a git command.
---------	--

ROOT CAUSE	Alias not set in git config, or typo in alias definition.
FIX	Add alias: <code>git config --global alias.st status</code> . List: <code>git config --global --list</code> .
	<pre>git config --global alias.st status git config --global alias.lg "log --oneline --graph --all" git config --global --list grep alias</pre>

#66 git log Showing Garbled Unicode Characters

SYMPTOM	git log shows ??? or garbled text instead of Unicode commit messages.
ROOT CAUSE	Terminal or pager (less) not configured for UTF-8 output.
FIX	Set LESSCHARSET=utf-8 and core.pager. Ensure terminal emulator uses UTF-8.
	<pre>git config --global core.pager 'less -r' export LESSCHARSET=utf-8 export LANG=en_US.UTF-8</pre>

#67 Credential Helper Not Caching Passwords

SYMPTOM	git push prompts for username/password every time despite entering it before.
ROOT CAUSE	Credential helper not configured; or cache timeout expired.
FIX	Configure credential helper: cache (Linux), osxkeychain (Mac), manager (Windows).
	<pre>git config --global credential.helper store # OR cache for 1 hour: git config --global credential.helper 'cache --timeout=3600'</pre>

#68 core.hooksPath Not Loading Custom Hooks

SYMPTOM	Custom pre-commit or commit-msg hooks not running; hooks directory configured but ignored.
ROOT CAUSE	core.hooksPath pointing to wrong directory, or hooks not executable.
FIX	Set correct path: <code>git config core.hooksPath .githubhooks</code> . <code>chmod +x</code> all hook files.
	<pre>git config core.hooksPath .githubhooks chmod +x .githubhooks/pre-commit git config core.hooksPath</pre>

#69 diff.tool Not Opening Configured Diff Tool

SYMPTOM	git difftool opens wrong tool; configured tool not launching.
ROOT CAUSE	diff.tool name doesn't match difftool.<name>.cmd definition, or tool not in PATH.
FIX	Configure both diff.tool name and difftool.<name>.cmd. Verify tool is in PATH.
	<pre>git config --global diff.tool vimdiff git config --global merge.tool vimdiff git difftool HEAD~1 HEAD</pre>

#70 init.defaultBranch Still Creating 'master'

SYMPTOM	git init creates 'master' branch instead of 'main'; new repos have inconsistent default.
ROOT CAUSE	init.defaultBranch not set in global git config; defaults to master in older Git.
FIX	Set globally: git config --global init.defaultBranch main.
	<pre>git config --global init.defaultBranch main git init # will now create main branch</pre>

CI/CD & WORKFLOWS — Real-World Troubleshooting

#71 CI Failing on Linting Due to Line Endings

SYMPTOM	CI lint step fails with CRLF/LF errors that don't appear locally (Windows dev, Linux CI).
ROOT CAUSE	Developer on Windows commits CRLF line endings; CI Linux runner expects LF.
FIX	Add .gitattributes with text=auto. Run git add --renormalize . to fix existing files.
<pre>echo '* text=auto eol=lf' > .gitattributes git add --renormalize . git commit -m "chore: normalize line endings"</pre>	

#72 Merge Queue / Protected Branch Blocking PR

SYMPTOM	PR cannot merge; branch protection requiring up-to-date branch with squash/rebase enforced.
ROOT CAUSE	Feature branch behind main after new commits; must be rebased before merge allowed.
FIX	Rebase feature branch onto latest main. Push with --force-with-lease.
<pre>git fetch origin git rebase origin/main git push --force-with-lease origin feature/branch</pre>	

#73 git bisect Finding Regression

SYMPTOM	A bug exists somewhere in 200 commits; need to identify which commit introduced it.
ROOT CAUSE	Regression introduced at unknown point in history; manual search impractical.
FIX	git bisect good/bad to binary-search commits. Automate with a test script.
<pre>git bisect start git bisect bad HEAD git bisect good v1.0.0 # Test each suggested commit, then: git bisect good # or bad git bisect reset</pre>	

#74 Pre-commit Hook Blocking All Commits

SYMPTOM	git commit always fails with hook error; unable to commit even clean code.
ROOT CAUSE	Pre-commit hook has a bug, dependency missing, or overly strict rule.
FIX	Debug hook: bash -x .git/hooks/pre-commit. Bypass temporarily: git commit --no-verify.
<pre>git commit --no-verify -m "emergency: bypass hook" # Fix hook then re-enable: bash -x .git/hooks/pre-commit</pre>	

#75 Repository Too Large — Slow Clone in CI

SYMPTOM	git clone in CI takes 3+ minutes; large repo history slowing pipelines.
---------	---

ROOT CAUSE	Accumulated large files, long history, or binary assets bloating repo size.
FIX	Use shallow clone in CI: <code>--depth=1</code> . Or <code>git sparse-checkout</code> for partial checkout.
	<pre>git clone --depth=1 --single-branch --branch main <url> # OR sparse checkout: git sparse-checkout init --cone git sparse-checkout set src/</pre>

#76 Commit Message Convention Failing CI Check

SYMPTOM	CI commit-lint step fails; conventional commit format not followed.
ROOT CAUSE	Commit message not following required format (e.g., feat:, fix:, chore: prefix).
FIX	Follow Conventional Commits spec. Set up commitlint locally to catch before push.
	<pre># Correct format: git commit -m "feat(auth): add OAuth2 login support" git commit -m "fix(api): handle null response from upstream"</pre>

#77 GitHub Actions Token Lacks Push Permission

SYMPTOM	CI workflow fails: remote: Permission to repo denied to github-actions[bot].
ROOT CAUSE	GITHUB_TOKEN permissions not set to write for contents in workflow.
FIX	Add permissions: contents: write to workflow YAML.
	<pre># In workflow YAML: # permissions: # contents: write git config user.email 'github-actions@github.com' git config user.name 'github- actions'</pre>

#78 Monorepo — CI Triggering on Unrelated Changes

SYMPTOM	Service A CI pipeline triggers when only Service B files changed.
ROOT CAUSE	CI not using path filters; all changes trigger all pipelines.
FIX	Add paths: or paths-ignore: filters to CI config to scope triggers per service.
	<pre># GitHub Actions: # on: # push: # paths: # - 'services/auth/**' # - 'shared/**'</pre>

#79 git worktree Causing Unexpected Conflicts

SYMPTOM	Using multiple worktrees; changes in one worktree affecting another.
ROOT CAUSE	Both worktrees share same .git directory; uncommitted changes or index conflicts.
FIX	Each worktree has independent index. Always commit or stash in one before switching.
	<pre>git worktree list git worktree add ../hotfix hotfix/v1 git worktree remove ../hotfix</pre>

#80 Changelog Not Generated Correctly From Commits

SYMPTOM	Auto-generated CHANGELOG missing entries or showing wrong commits.
ROOT CAUSE	Commits not following conventional format, or tag range incorrect for changelog generation.
FIX	Use standard-version or git-cliff with correct tag range. Fix commit messages going forward.
	<pre>git log v1.0.0..HEAD --oneline --no-merges npx standard-version --dry-run</pre>

ADVANCED GIT — Real-World Troubleshooting

#81 git filter-repo Removing File From Entire History

SYMPTOM	Sensitive file or large binary needs to be completely removed from all historical commits.
ROOT CAUSE	File committed by mistake and needs purging from all history for security or size.
FIX	git filter-repo --path <file> --invert-paths. Then force push all branches.
<pre>git filter-repo --path secrets.env --invert-paths git push origin --force --all git push origin --force --tags</pre>	

#82 Repository Corruption — Object Database Error

SYMPTOM	git status or git log fails: object file is empty or loose object is corrupt.
ROOT CAUSE	Disk error, interrupted write, or filesystem corruption damaged git object store.
FIX	git fsck to find corrupt objects. Restore from clone or remote.
<pre>git fsck --full git clone --mirror origin backup-repo git remote update</pre>	

#83 git gc Making Repository Slower Temporarily

SYMPTOM	After git gc, operations are slower or repo seems broken temporarily.
ROOT CAUSE	gc running in background; pack files being rewritten; concurrent access conflict.
FIX	Run git gc --aggressive during off-hours. Wait for completion. git fsck after.
<pre>git gc --aggressive --prune=now git count-objects -vH</pre>	

#84 Partial Clone Not Fetching Needed Objects

SYMPTOM	After partial clone (--filter=blob:none), operations fail: missing object.
ROOT CAUSE	Partial clone lazily fetches blobs; some operations require objects not yet fetched.
FIX	Fetch missing objects: git fetch --filter=blob:none. Or convert to full clone.
<pre>git fetch --filter=blob:none origin git cat-file -e <sha> # check object exists</pre>	

#85 git rebase -i Accidentally Dropping Commits

SYMPTOM	After interactive rebase, commits are missing from branch history.
ROOT CAUSE	Lines deleted or marked 'drop' in rebase todo list inadvertently.

FIX	git reflog to find original branch tip before rebase. Reset branch to that commit.
	<code>git reflog head -20 git reset --hard HEAD@{5} # before rebase</code>

#86 Merge Strategy — Ours vs Theirs Confusion

SYMPTOM	After merge with -s ours, expected changes from other branch are missing.
ROOT CAUSE	-s ours strategy keeps current branch content entirely; discards other branch changes.
FIX	Use -X ours (extended option) to prefer current branch only in conflicts, not discard all.
	<code>git merge -X ours feature/branch # prefer ours on conflict # vs git merge -s ours feature/branch # discard theirs entirely</code>

#87 git notes Not Preserved After Rebase

SYMPTOM	git notes attached to commits are lost after rebasing or cherry-picking.
ROOT CAUSE	Rebase creates new commit SHAs; notes reference old SHAs that no longer exist on branch.
FIX	Use git notes copy to migrate notes to new SHAs. Or use commit trailers instead.
	<code>git notes copy <old-sha> <new-sha> git notes show <commit></code>

#88 Alternates Object Store Broken

SYMPTOM	git fails with: object file does not exist after moving repository or using alternates.
ROOT CAUSE	Repository configured to share objects via .git/objects/info/alternates; path broken.
FIX	Fix path in .git/objects/info/alternates or run git repack to consolidate objects.
	<code>cat .git/objects/info/alternates git repack -a -d --no-reuse-delta</code>

#89 git archive Missing Files

SYMPTOM	git archive output missing some files that exist in repository.
ROOT CAUSE	Files excluded by export-ignore attribute in .gitattributes.
FIX	Check .gitattributes for export-ignore. Remove if file should be in archives.
	<code>grep export-ignore .gitattributes git archive --format=tar HEAD tar -t grep <filename></code>

#90 Ref Namespace Collision

SYMPTOM	git branch name conflicts with existing tag or remote ref; operations behave unexpectedly.
----------------	--

ROOT CAUSE	Branch and tag with same name; git ambiguity in ref resolution.
FIX	Use full refspec: refs/heads/<branch> vs refs/tags/<tag> to disambiguate.
	<pre>git show refs/heads/v1.0 git show refs/tags/v1.0 git branch -d refs/heads/v1.0</pre>

#91 Broken git Hooks After Repository Clone

SYMPTOM	Hooks in .git/hooks/ don't run after fresh clone; team hooks not shared.
ROOT CAUSE	.git/hooks are not tracked by git; cloning doesn't bring hooks.
FIX	Store hooks in tracked directory (e.g., .githooks). Configure core.hooksPath.
	<pre>git config core.hooksPath .githooks chmod +x .githooks/*</pre>

#92 Repository Migrated — Refs Not Updated

SYMPTOM	After moving repo to new remote, git push fails or points to old remote.
ROOT CAUSE	Old remote URL still set; remote tracking branches reference old host.
FIX	Update all remotes: git remote set-url. Fetch from new remote to update tracking.
	<pre>git remote set-url origin https://newhost.com/org/repo.git git fetch --all git branch -u origin/main main</pre>

#93 git log --all Not Showing Expected Branch

SYMPTOM	git log --all or gitk missing a branch that should exist.
ROOT CAUSE	Branch exists as remote-tracking only; not in local refs/heads.
FIX	Create local tracking branch: git checkout -b <branch> origin/<branch>.
	<pre>git branch -r # list remote branches git checkout -b feature/x origin/feature/x</pre>

#94 Pack File Corruption After Interrupted Push

SYMPTOM	After network interruption during push/fetch, git operations fail with pack file errors.
ROOT CAUSE	Partial pack file written to .git/objects/pack during interrupted transfer.
FIX	Remove incomplete pack files from .git/objects/pack/. Run git fsck.
	<pre>ls .git/objects/pack/ git fsck --full git remote update</pre>

#95 git sparse-checkout Not Excluding Directories

SYMPTOM	git sparse-checkout configured but all files still checked out.
---------	---

ROOT CAUSE	Sparse checkout not enabled, or cone mode patterns not matching directory structure.
FIX	Enable sparse checkout: <code>git sparse-checkout init --cone</code> . Set paths explicitly.
	<pre>git sparse-checkout init --cone git sparse-checkout set src/ docs/ git sparse-checkout list</pre>

#96 Bundle File Cannot Be Imported

SYMPTOM	git fetch from bundle file fails: does not contain necessary objects.
ROOT CAUSE	Bundle missing prerequisite commits; target repo doesn't have base commits bundle was created from.
FIX	Create bundle with full history: <code>--all</code> flag. Or ensure base commit exists in target.
	<pre>git bundle create full-repo.bundle --all git bundle verify full-repo.bundle git clone full-repo.bundle new-repo</pre>

#97 git replace Causing Confusing History

SYMPTOM	git log shows unexpected commit content; history seems inconsistent between clones.
ROOT CAUSE	git replace refs grafting different commit content onto existing SHAs; not pushed by default.
FIX	List replacements: <code>git replace -l</code> . Remove: <code>git replace -d <sha></code> . Push: <code>git push origin refs/replace/*</code> .
	<pre>git replace -l git replace -d <original-sha> git push origin 'refs/replace/*'</pre>

#98 Shallow Clone Causing rebase/bisect Failures

SYMPTOM	git rebase or git bisect fails: shallow repository... needs unshallow.
ROOT CAUSE	Shallow clone doesn't have full commit ancestry needed for rebase or bisect.
FIX	Unshallow the clone: <code>git fetch --unshallow</code> .
	<pre>git fetch --unshallow git fetch --deepen=100 # partial unshallow</pre>

#99 Repository Growing Too Large — Identifying Bloat

SYMPTOM	Repository size is unexpectedly large; clone and fetch are slow.
ROOT CAUSE	Large binary files, accidentally committed node_modules/build artifacts in history.
FIX	Use git-sizer or git rev-list to find large objects. Remove with filter-repo.
	<pre>git rev-list --objects --all git cat-file --batch-check='% (objecttype) %(objectname) %(objectsize) %(rest)' sort -k3 -rn head -20</pre>

#100 git maintenance Not Running Automatically

SYMPTOM	Repository gradually slowing down; pack files accumulating; no auto-maintenance.
ROOT CAUSE	git maintenance schedule not configured; background optimization not running.
FIX	Enable background maintenance: git maintenance start. Configures OS-level scheduler.
<pre>git maintenance start git maintenance run --task=gc git maintenance stop</pre>	